The essential lesson is not that everyone should be shipping fifty times per day but that by reducing batch size, we can get through the Build-Measure-Learn feedback loop more quickly than our competitors can. The ability to learn faster from customers is the essential competitive advantage that startups must possess.

## SMALL BATCHES IN ACTION

To see this process in action, let me introduce you to a company in Boise, Idaho, called SGW Designworks. SGW's specialty is rapid production techniques for physical products. Many of its clients are startups.

SGW Designworks was engaged by a client who had been asked by a military customer to build a complex field x-ray system to detect explosives and other destructive devices at border crossings and in war zones.

Conceptually, the system consisted of an advanced head unit that read x-ray film, multiple x-ray film panels, and the framework to hold the panels while the film was being exposed. The client already had the technology for the x-ray panels and the head unit, but to make the product work in rugged military settings, SGW needed to design and deliver the supporting structure that would make the technology usable in the field. The framework had to be stable to ensure a quality x-ray image, durable enough for use in a war zone, easy to deploy with minimal training, and small enough to collapse into a backpack.

This is precisely the kind of product we are accustomed to thinking takes months or years to develop, yet new techniques are shrinking that time line. SGW immediately began to generate the visual prototypes by using 3D computer-aided design (CAD) software. The 3D models served as a rapid communication tool between the client and the SGW team to make early design decisions.

The team and client settled on a design that used an advanced